

Germans Savcisens, PhD

Postdoctoral Research Associate, Northeastern University

Boston, MA, USA · [Portfolio](#) · [LinkedIn](#) · [GitHub](#) · [Google Scholar](#) · [ORCID](#) · [Semantic Scholar](#) · [Scopus](#) · [Web of Science](#)

PROFILE

Postdoctoral researcher (PhD, Applied Mathematics and Computer Science) specializing in statistical and machine-learning modeling of registry-scale health & behavior data, and in the robustness, and stability assessment of AI systems. Investigated transformer-based systems for nation-scale health/socioeconomic trajectory prediction ([life2vec](#), trained on national registry and electronic health records), veracity probing of LLMs and multi-agent LLM simulation (CoevolveSim), with end-to-end Python experience from data extraction and cleaning to modeling and interpretability analysis. Works across multidisciplinary teams and mentors graduate researchers in interpretable ML and AI auditing.

Areas of interest: population-scale health & behavior modeling, explainable AI & AI auditing, bias & robustness assessment, transformer architectures, agent-based simulation, evaluation of predictive models.

EDUCATION

Technical University of Denmark

2020–2024

PhD, Applied Mathematics and Computer Science

Copenhagen, Denmark

- Dissertation: *Life Trajectories as Symbolic Language: Exploring Human Behaviour with Language Models*
- Committee: Søren Hauberg (chair), Mads Nielsen, Arkadiusz Stopczynski

Technical University of Denmark

2018–2020

MSc, Human-Centered Artificial Intelligence

Copenhagen, Denmark

- Thesis (Collaboration with Novo Nordisk): *Analyzing Health Data Records Using Neural Networks*
- Thesis Advisors: Sune Lehmann (DTU) and Adam Lenart (Novo Nordisk)

Aalto University

2021

Visiting Student, Digital Ethics & Statistical Natural Language Processing

Helsinki, Finland

Aalborg University

2013–2016

BSc, Human-Computer Interaction (Medialogy)

Copenhagen, Denmark

RESEARCH EXPERIENCE

Northeastern University

May 2024–present

Postdoctoral Research Associate | Mentor: Tina Eliassi-Rad

Boston, USA

- Developed LLM probing frameworks for knowledge representation and belief stability [[sAwMIL](#)].
- Designed a multi-agent framework (using R and Python) to study opinion dynamics in networks of LLM agents [[CoevolveSim](#)].
- Supervised graduate researchers in interpretable machine learning and LLM auditing; coordinated research on multi-agent interactions.
- Presented projects at the NeurIPS Mechanistic Interpretability Workshop, IC2S2, and NEMI Workshop.

Technical University of Denmark & University of Copenhagen

Sep 2020–Apr 2024

PhD Student & Research Assistant | Advisors: Sune Lehmann, Lars Kai Hansen

Copenhagen, Denmark

- Led modeling of nation-scale socioeconomic and health trajectories using transformer-based models [[life2vec](#)].
- Developed and evaluated life-outcome prediction models, with emphasis on interpretability.
- Taught and developed course materials for a graduate-level course in Social Network Science.
- Invited talks at MIT, University of Oxford, NordicAI and Complexity Science Hub.

Statistics Denmark, Data Science Lab

Sep 2020–Apr 2024

Guest Researcher

Copenhagen, Denmark

- Collaborated with economists and social scientists to conduct quantitative and qualitative analyses of demographic and socioeconomic patterns in observational registry data.
- Led modeling of large-scale longitudinal datasets.

Network Science Institute, Northeastern University

Sep 2022–Feb 2023

Visiting PhD Student

Boston, USA

- Analyzed ethical and societal risks of deep learning applied to socioeconomic data.
- Investigated methods for enhancing transparency and fairness in algorithmic systems.

Novo Nordisk, R&D (Real World Evidence)

Sep 2019–Apr 2020

Researcher (Student Assistant)

Copenhagen, Denmark

- Conducted research on predictive models for medical adherence using large-scale Electronic Health Records (EHRs), addressing missing data and selection bias in the observational cohort.
- Developed hierarchical attention-based neural models for health outcome prediction.

- Led preprocessing and integration of behavioral and clinical data sources.

PUBLICATIONS

PEER REVIEWED & PUBLISHED

- G. Savcicens** & T. Eliassi-Rad (2026). Trilemma of Truth in Large Language Models. **To appear** in *Proceedings of the 2026 Conference on Empirical Methods in Natural Language Processing*. [\[arXiv\]](#) [\[Code\]](#) [\[NeurIPS Poster\]](#)
- S. Dies, C. Maynard, **G. Savcicens** & T. Eliassi-Rad (2026). Epistemic Familiarity is Associated With Belief Stability in Large Language Models. **To appear** in *Findings of the 2026 Conference on Empirical Methods in Natural Language Processing*. [\[arXiv\]](#)
- G. Savcicens** (2025). Large language models act as if they are part of a group. *Nature Computational Science*. [\[DOI\]](#)
- Z. Shafi, **G. Savcicens** & T. Eliassi-Rad (2025). REGE: A method for incorporating uncertainty in graph embeddings. *SIAM International Conference on Data Mining*. [\[DOI\]](#)
- E. Denove, E. Michelet, **G. Savcicens** & E. F. Fernández (2024). An Industrial West? A Mixed-Methods Analysis of Newspapers Discourses about Technology. *Journal of Data Mining & Digital Humanities*. [\[DOI\]](#)
- E. F. Fernández & **G. Savcicens** (2024). Quantifying Western Discourses about Technology. *Digital Humanities in the Nordic and Baltic Countries*. [\[DOI\]](#)
- G. Savcicens**, T. Eliassi-Rad, L. Kai Hansen, L.H. Mortensen, L. Lilleholt, A. Rogers, I. Zettler & S. Lehmann (2024). Using sequences of life-events to predict human lives. *Nature Computational Science*. [\[DOI\]](#) [\[Official Page\]](#)
- E. F. Fernández & **G. Savcicens** (2023). A Sustainable West? Analyzing Clusters of Public Opinion in Sustainability. *Digital Humanities in the Nordic and Baltic Countries*. [\[DOI\]](#)

PREPRINTS

- M. Laber, Z. Shafi, **G. Savcicens**, M. Chinazzi, S. V. Scarpino, A.-L. Barabási & T. Eliassi-Rad (2026). Scaling of Capability and Efficiency at Inference Time in Large Reasoning Models. *arXiv*. [\[arXiv\]](#)
- G. Savcicens**, S. Dies, C. Maynard & T. Eliassi-Rad (2026). Belief coevolution in a social network of generalist and specialist large language models. *arXiv*. [\[arXiv\]](#) [\[Code\]](#) [\[NetSci Poster\]](#)
- E. Sivak,...,**G. Savcicens**, et al. (112 authors) (2026). Births are difficult to predict even with rich survey and full-population register data. *arXiv*. [\[arXiv\]](#)

BLOGPOST

- G. Savcicens** (2026). Is Life2vec a Mess? What's Wrong with AI Death Calculators? *Personal Page*. [\[Source\]](#)

THESES

- G. Savcicens** (2023). Life Trajectories as Symbolic Language: Exploring Human Behaviour with Language Models. *PhD Thesis*. [\[DTU\]](#)
- G. Savcicens** (2020). Analyzing Health Data Records Using Neural Networks: Predicting Diabetes-Related Clinical Outcomes. *MSc Thesis*. [\[DTU\]](#)

TEACHING & MENTORING

Northeastern University	2024–2026
<i>Guest Lecturer</i> <i>Foundations of AI</i> (2026); <i>Machine Learning with Graphs</i> (2024 & 2025)	
IT University of Copenhagen	2022–2023
<i>Principal Lecturer</i> (2023); <i>Assistant Lecturer</i> (2022) <i>Algorithmic Fairness, Accountability and Ethics</i>	
Technical University of Denmark	2019–2021
<i>Teaching Assistant</i> <i>Advanced Machine Learning</i> ; <i>Social Data Analysis</i> ; <i>Social Graphs and Interactions</i>	
Mentoring & Supervision	
Co-supervised 5 MSc thesis and term projects on NLP for legal/compliance documents and longitudinal app-usage modeling, Technical University of Denmark (2021–2024).	

SELECTED TALKS & PRESENTATIONS

Interpretability or Guesswork: What Happens When We Open Up Large Language Models	Sep 2026
<i>International Conference on Social Computing (Keynote Speaker)</i>	Oxford, UK
Transformer Architectures for Computational Social Science [Page]	Sep 2026
<i>International Conference on Social Computing (Workshop Lead)</i>	Oxford, UK
Belief Coevolution in a Social Network of LLMs	Aug 2026
<i>New England Mechanistic Interpretability Workshop (Poster)</i>	Boston, USA
Belief Coevolution in a Social Network of LLMs	Jun 2026
<i>International Conference on Network Science (Lightning Talk)</i>	Boston, USA
Formal Epistemology of Large Language Models	Apr 2026
<i>Social Complexity Lab, University of Copenhagen (Talk)</i>	Copenhagen, Denmark

From Human Lives to Machine Minds [Slides] <i>Rising Stars in Computational & Data Science, Santa Fe Institute (Talk)</i>	Apr 2026 Santa Fe, USA
The Trilemma of Truth in LLMs <i>NeurIPS Mechanistic Interpretability Workshop (Poster)</i>	Dec 2025 San Diego, USA
Life2vec: Foundation Models for Registry Data [Slides] <i>Networks for Healthy Aging Workshop (Invited Speaker)</i>	Nov 2025 Vienna, Austria
From Life-Courses to Representations: Foundation Models for Registry Data <i>SWECOV Workshop (Keynote Speaker)</i>	Sep 2025 Stockholm, Sweden
The Trilemma of Truth in Large Language Models <i>New England Mechanistic Interpretability Workshop (Poster)</i>	Aug 2025 Boston, US
Improving Probes that Track Veracity in LLMs <i>Conference on Computational Social Science (Poster)</i>	Jul 2025 Norrköping, Sweden
Workshop on the '(Un)predictability in the Social Sciences' <i>University of Helsinki (Invited Speaker)</i>	Feb 2025 Helsinki, Finland
LLMs for Health and Labor Data <i>Max Planck Institute (Invited Speaker)</i>	Jan 2025 Rostock, Germany
Life2vec: Life Trajectories in High-dimensional Spaces [Recording] <i>Chinese University of Hong Kong (Invited Speaker)</i>	Nov 2024 Hong Kong, HK (Remote)
Life2vec: Life Trajectories in High-dimensional Spaces (Invited Speaker) <i>GenLife</i>	Apr 2024 Copenhagen, Denmark
Seminar Talk on Life2vec <i>University of Oxford (Invited Speaker)</i>	Jan 2024 Oxford, UK (Remote)
Learning Representations of Socio-economic Data <i>ODISSEI Lecture (Invited Speaker)</i>	Sep 2023 Netherlands (Remote)
Danish Registry Data and Life Trajectories <i>Human Dynamics Lab (MIT)</i>	Oct 2022 Boston, US

PROFESSIONAL SERVICE

Senior Program Committee: AAAI (Special Track on AI Alignment) 2027.

Session Chair: International Conference on Network Science (ML & AI track) 2026.

Journals (reviewer): Artificial Intelligence Review, EPJ Data Science, Humanities and Social Sciences Communications, Nature Computational Science, npj Complexity, Science Advances, Scientific Reports.

Conferences & Workshops (reviewer): NeurIPS (Ethics Reviewer 2023–2026; AXIOM Workshop 2026; Mechanistic Interpretability Workshop 2025), Workshop on NLP and Computational Social Science 2026, IC2S2 2025, Digital Humanities in the Nordic and Baltic Countries 2025, Conference on Complex Networks and Their Applications 2025, EMNLP 2024.

AWARDS & GRANTS

Rising Stars in Computational and Data Sciences 2026 <i>Santa Fe Institute</i>	2026 USA
Travel Grants <i>Otto Mønsted Fond, STIBO Foundation, William Demant Fonden</i>	2022 Denmark